Industrial Robot Adoption and Employment in European Manufacturing: The Role of Collective Bargaining Coordination

Alice Andersson (50818) and Erik Gunnarsson (24928)
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Master in International Business
Stockholm School of Economics

Style Guide:

Only Black/white text color (gdocs can use grey scale as a default sometimes)

Title (Rubrik)

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1. Introduction

Concerns about labour-displacing technology date back to the nineteenth-century Luddites, who destroyed textile machinery they believed threatened their livelihoods (Autor, 2015; Mokyr et al., 2015; Encyclopaedia Britannica). Such the same type of concern remain today, the technological context has fundamentally changed. The stock of industrial robots in European countries quadrupled between 2000 and 2017 (Bachmann et al., 2024), and their capabilities have expanded rapidly, enabling them to perform an increasingly wide range of tasks with minimal human intervention (Graetz and Michaels, 2017). As robots have become more capable of replacing humans, the consequences for employment have become arguably more salient.

Empirical evidence on these consequences, however, are divided. For example, influential studies from the United States find that robot adoption reduces employment (Acemoglu and Restrepo, 2020) while evidence from Germany, which has a considerably larger robot stock per capita compared to the U.S, find no reduction in employment (Dauth et al, 2021). This divergence is not limited to transatlantic comparisons; significant heterogeneity exists across European countries (Bachmann et al., 2022). This variation raises a central question: why does exposure to the same technology generate different employment outcomes across countries?

Research finds that labor market institutions play an important role in this question. While prior studies have examined union density, defined as the share of union members out of wage earners (Parolin, 2020; Haapanala et al., 2022), this approach has been criticized as overly simplistic. What appears to matter more is the degree of coordination across bargaining units (Garnero, 2021; Bhuller, 2022; Leibrecht et al., 2023) referring to the extent to which unions and employers coordinate wage-setting across sectors, industries, or bargaining levels rather than negotiating independently.

This notion is reflected by the Hawk–Dove framework (Bhuller et al., 2022; Leibrecht et al., 2023), positing that employment effects of automation depend on how collective bargaining systems are coordinated. In coordinated systems, unions negotiate across sectors and internalize the broader macroeconomic consequences of their demands. Such institutional structures are predicted to facilitate adjustment to technological change. In uncoordinated systems, unions represent narrower groups and do not internalize economy-wide spillovers, potentially amplifying displacement pressures.

If collective bargaining shapes how labour markets respond to automation, identifying the underlying mechanism is of both scientific and policy relevance. Scientifically, it helps explain why the employment effects of automation differ across European countries, as existing research has identified this variation but provides limited evidence on the institutional mechanisms behind it (Leibrecht et al., 2023). From a policy perspective, clarifying this mechanism informs how governments can capture productivity gains from automation while mitigating displacement risks. This is particularly timely, as collective bargaining, while central to labour market adjustment, is highly pressured due to the long-standing decline in trade union density and bargaining coverage (OECD, 2019).

Hence, the purpose of this study is to examine whether the manufacturing employment effects of industrial robot exposure depend on the structure of collective bargaining. Europe provides a suitable empirical setting, as it combines substantial institutional variation with a relatively comparable economic environment and harmonized cross-country data that enable industry-level analysis.

2. Literature Review

2.1 Robotics and employment in manufacturing

The potential for technological progress to generate technological unemployment has long been a central concern in labor economics, with industrial robots attracting particular attention due to their unique capacity for labor substitution (Hötte et al., 2023; Dauth et al, 2021) Indeed, this capacity is reflected in their very definition: automatically controlled, reprogrammable, and multipurpose manipulators programmable in three or more axes for use in industrial automation applications (IFR;ISO 8373:2021). Because they fundamentally are a labor-saving technology, their adoption alters the task content of production through a displacement effect, where capital directly takes over tasks previously performed by labor. Consequently, the implementation of industrial robots exerts a significant downward pressure on employment and wages in robot-exposed industries, all else being equal (Autor & Salomons, 2018;Acemoglu & Restrepo, 2019)

However, the empirical evidence regarding the net employment effects of robots reveals stark contradictions. In the United States, research shows large and robust negative effects of robots on local employment (Acemoglu and Restrepo 2020)) Conversely, studies focusing on European economies often observe no aggregate employment decline (Graetz and Michaels, 2018; Lábaj and Vitáloš, 2020). This has given rise to a theoretical: the “techno-pessimist” perspective emphasizing job displacement and labor market exclusion versus the “techno-optimist” perspective focuses on productivity gains, task reinstatement, and market expansion (Aghion et al., 2022; Lamperti and Castellani, 2026). However, in spite of this divergence, there is a broad agreement that the effects of automation are heterogeneous across groups, including worker skills, demographics, and industries

(Roine, 2023; Hötte et al, 2023). To fully understand the impact of robots on employment, there is a need for deeper analysis beyond the macro-level (Otto and Abraham, 2025).

The manufacturing sector is found to be particularly exposed to labour replacement, as its highly routinized production processes make it the primary environment for industrial robot adoption (Filipi et al, 2023). Several studies have found reduced employment in this sector (Mann and Puttmann, 2018; Dauth et al, 2021; xxx?) However, even within manufacturing, the net employment effects of robots remain highly ambiguous due to conflicting dynamics (Aghion et al, 2022 [right citation?]). On one hand, automation can lead to job displacement, with evidence showing that exogenous exposure to robots reduces manufacturing employment and increases the probability of incumbent workers separating from their employers (Bessen et al, 2025; Bonfiglioli et al, 2024). On the other hand, automating firms often experience productivity gains that allow them to lower prices and expand their market share at the expense of non-automating competitors, creating new jobs (Koch et al, 2019; Aghion et al, 2022).

Ultimately, the heterogeneous outcomes observed across advanced economies suggest that robotization alone does not deterministically dictate employment trajectories (Doorley et al, 2024; Haapanala et al, 2022). Rather, the extent to which labor-saving technologies translate into job displacement or productivity-driven employment gains is moderated by the institutional environment in which firms operate. In particular, labour market institutions play a critical role in absorbing the technological shock (Leibrecht et al, 2023).

2.2 Robotics, employment and collective bargaining

A particular labour market institution is collective bargaining. Based on a complex set of rules and practices, partly written in national laws and partly based on longstanding traditions, collective bargaining is the process through which workers and employers associate to express their interests and bargain over the terms and conditions of employment (OECD, 2019). Because new technology fundamentally alters the organization of work, its implementation is subject to these bargaining processes (Berge and Dekker, 2019). Consequently, the employment effects of technological change depend partly on workers' bargaining power to pressure management to mitigate unemployment. Empirical evidence supports this effect: in a U.S. context, union membership slows displacement in routine-intensive occupations, reducing technological unemployment by strengthening workers' bargaining power (Parolin, 2020). Further, comparing OECD countries, workers in countries of higher union density report significantly less fear of robotization (Dekker, 2017).

However, the literature also emphasizes that strong collective bargaining does not universally yield positive employment outcomes. First, union wage policies can inadvertently accelerate technological displacement (Kostøl and Svarstad, 2023). Because unions typically fight for wage compression, thereby raising the relative wages of routine workers to non-routine workers, they may cause increasing labour costs relative to capital and a decline in the demand for labour demand. Second, when adapting to these technological shocks, trade unions often act to protect their core constituents rather than the broader workforce, fostering labor market dualization. Empirical evidence shows that unions prioritize the job security of insiders, typically incumbent workers, at the expense of outsiders, such as new labour market entrants, younger, and lower-educated workers, shifting the burden of technological displacement onto these groups (Dauth et al., 2021; Haapanala et al., 2023). Other

research, however, does not find evidence for the insider-outsider theory, arguing that in coordinating collective bargaining systems, all workers are protected (Garnero, 2020; Leibrecht, 2023).

While collective bargaining remains a central labour market institution, trade union membership has experienced a steady decline across advanced economies over the last few decades (OECD, 2019; Bhuller et al, 2022). This decline in union density might seem to suggest that unions have lost their protective power, but union density alone understates institutional influence. This is because, firstly, the share of workers covered by the terms of collective bargaining agreements often vastly exceeds actual union membership (ibid). Secondly, because what matters is the 'entire package'; how these institutions are coordinated (Garnero, 2020). While this distinction matters little in the United States, in many European countries, this widespread coverage is rooted in national statutes and practices that extend collective agreements to non-union workers, firms, and sectors (Bhuller et al, 2022).This explains why empirical evidence finds no association between the level of unionization and workers' expectations regarding the effects of new technology (ten Berge and Dekker, 2025).

Leibrecht el al (2023) applies these findings to the context of robotisation. By explicitly distinguishing trade union strength from bargaining coordination, the authors demonstrate that increased automation is associated with higher unemployment in countries with uncoordinated bargaining systems, such as the United States. Conversely, this negative association is significantly muted in coordinated systems, such as Germany. Their results confirm that employment outcomes following automation depend not merely on the strength of labor institutions, but fundamentally on how they are structurally organized.

2.3 Research Gap and Research Question

The question of whether robots will replace human jobs has long occupied researchers, yet no clear consensus has emerged. What is found, however, is that countries experience automation differently depending on their labour market institutions. Collective bargaining has been shown to moderate the relationship between automation and employment, although evidence on its direction remains mixed. Recent research further suggests that the coordination of collective bargaining structures is more decisive for employment outcomes than union density alone. However, with the exception of Leibrecht et al. (2023), the specific role of bargaining coordination in moderating the relationship between industrial robot adoption and employment remains largely unexplored. Importantly, their analysis relies on aggregated macroeconomic data capturing net employment effects. Recognizing the need for more granular analysis, Leibrecht et al. (2023) explicitly call for future research examining the dynamics of automation, employment and collective bargaining at the industry level. Given that manufacturing is particularly exposed to automation technologies, focusing on this sector provides a relevant setting to assess how bargaining coordination shapes employment responses. x

This motivates the following research question:

To what extent does collective bargaining coordination influence the relationship between industrial robot adoption and employment in European manufacturing industries?

3. Theoretical Framework

3.1 The Task-Based Framework

To conceptualize the impact of industrial robots on manufacturing employment, this thesis draws on the Task-Based Framework (Acemoglu and Restrepo, 2018, 2019; Acemoglu and Autor, 2011; Autor, Levy, and Murnane, 2003); Zeira (1998). This framework posits that production requires tasks, which are allocated to capital or labor. It also posits that automation technologies impact the allocation of tasks between these two factors of production; they change the *task content of production*. By doing so, automation technologies impact employment through three competing effects:

- *The displacement effect,* occurring as automation substitutes human labour in specific tasks and reduces the labour share of value added.

- *The productivity effect,* occurring when automation reduces production costs and increases value added, enables firms to expand output and raise the demand for labor in remaining, non-automatable tasks.

- *Reinstatement effect*, referring to the creation of new labor-intensive tasks in which humans retain a comparative advantage. Being the opposite of the replacement effect, this increases overall labor demand and the labor share of income.

The net impact on employment depends on which effect dominates. However, these economic mechanisms do not operate in a vacuum; they are filtered through labour market institutions. x

3.2 The Hawk-Dove Divide

This thesis focuses on collective bargaining coordination as a labour market institution, conceptualized through the Hawk–Dove Divide. Originating in classical game theory and applied to collective bargaining by Bhuller et al. (2022), this theory predicts that bargaining behaviour is dictated by how unions are coordinated. Coordination operates along two dimensions: horizontally, referring to alignment of bargaining across different sectors, and vertical coordination, referring to whether bargaining takes place at the firm, industry, or national level. Higher levels of coordination are associated with more cooperative ("dovish") union behaviour, while lower levels are associated with more conflictual ("hawkish") union behaviour, with implications for labour market outcomes.

More recently, Leibrecht et al. (2023) applied this theory to technological change. The authors demonstrated how unions respond to the threat of industrial robots, and ultimately how robot adoption translates into employment effects, is moderated by the coordination of countries' collective bargaining systems.

Coordinated Collective Bargaining Systems

In coordinated systems, unions represent a broad, interdependent workforce who are synchronized in their demands. Hence, they recognize that aggressively pushing for higher wages for one group could trigger price increases or job losses for their complementary members elsewhere. Consequently, they

are forced to internalize these macroeconomic costs and behave “dovishly”. When faced with robot adoption dovish unions accept automation as a structural adjustment and practice wage restraint. By deliberately keeping human labor cost-competitive relative to the new machinery, they reduce the firm’s financial incentive to replace workers with robots, effectively dampening job displacement and supporting long-run employment.

Uncoordinated Systems

In uncoordinated systems, unions represent narrow groups of workers. Because they do not have to consider the economic fallout on other sectors, they do not internalize broader costs like aggregate job loss. Instead, they behave “hawkishly” by aggressively pursuing wage premiums to protect their specific incumbent members. However, this hawkish behavior backfires when facing automation: by driving up the cost of human labor relative to capital, they inadvertently strengthen a firm’s incentive to invest in robots, ultimately accelerating job displacement and increasing overall unemployment.

Furthermore and crucially, the Hawk-Dove Divide argues that bargaining structure matters significantly more than raw union density alone (Bhuller et al., 2022). This helps explain why collective bargaining continues to powerfully shape employment outcomes despite the steady decline in formal union membership over recent decades (OECD, 2019).

x see Appendix 1 for game theory

3.3 Summary of Hypotheses

Based on these frameworks, this thesis posits that:

H1: Industrial robot adoption generates a displacement effect in manufacturing employment, but the net magnitude of this job loss exhibits significant cross-country heterogeneity.

H2: The relationship between industrial robot adoption and unemployment is not significantly moderated by union density.

H3: In coordinated collective bargaining systems, unions act as "Doves" by internalizing macroeconomic externalities, muting the positive association between robot adoption and unemployment.

H4: In uncoordinated collective bargaining systems, unions act as militant "Hawks" which raises the relative cost of human labor, accelerating substitution and exacerbating the positive association between robot adoption and unemployment.

4. Data Collection

4.1 Research Design Overview

This study uses a quantitative, observational, longitudinal panel design at the country-industry-year level. The panel covers EU manufacturing countries over the years 2000–2014 (the WIOD coverage window), with institutional moderators (coordination, union density, and adjusted coverage) frozen at pre-period baselines to avoid reverse causality.

4.2 Data Sources

4.2.1 Robot Adoption (IFR)

Our core measure of robot adoption comes from the International Federation of Robotics (IFR), using a dataset covering 1993–2019 provided by XXX. The data provide annual operational stock figures for industrial robots by country and manufacturing industry, defined according to ISO 8373 standards. The operational stock represents robots currently in use, adjusted for depreciation and retirement, rather than cumulative installations, providing a more accurate reflection of productive robots at work.

Robot intensity is constructed as robots per thousand workers, with the denominator fixed at a baseline year, following Leibrecht et al., (2023) to ensure the measure captures year-on-year changes in robot stocks rather than workforce fluctuations. Robot intensity is lagged one period in all specifications to reduce simultaneity concerns.

4.2.2 Employment Outcomes (WIOD)

Our primary outcome variable is industry-level gross exports, drawn from the World Input-Output Database (WIOD, 2016 release; Timmer et al., 2015). WIOD provides harmonised international trade data in the form of inter-country input-output tables, covering 43 countries and 56 (ISIC Rev. 4[1]) industries from 2000–2014.

We extract each country's industry-level gross exports (the total value of goods shipped to all foreign destinations) from the WIOD trade tables. Exports are expressed in current US dollars and log-transformed in all specifications. This measure captures the competitiveness channel through which robot adoption may affect industry outcomes: automation-driven productivity gains lower unit costs, improving price competitiveness in international markets.

4.2.3 Industry-Level Controls (WIOD SEA)

To control for time-varying industry characteristics, we draw from WIOD Socio-Economic Accounts (WIOD SEA). The SEA provides industry-level data on value added, capital compensation, labour compensation, employment, and hours worked, harmonised to match the WIOD industry classification.

[1] International Standard Industrial Classification of All Economic Activities (ISIC), Rev.4

We include two industry-level controls sourced from the SEA:

Log value added (ln_va): captures industry size and productivity, controlling for the possibility that larger or more productive industries trade more and adopt more robots simultaneously.

Log capital stock/compensation (ln_cap): controls for non-robot capital intensity, ensuring that the robot adoption coefficient is not confounded by broader capital deepening.

4.2.3 Labor Market Institutions (ICTWSS)

Institutional moderators are taken from the ICTWSS database (Institutional Characteristics of Trade Unions, Wage Setting, State Intervention and Social Pacts), which supplies a comprehensive annual data on labour market institutions across OECD countries from 1960 onward. We initially used three moderators:

Wage coordination (coord): As the Primary moderator, ICTWSS coordination index ranges from 1 (fragmented firm-level bargaining) to 5 (maximum centralised or government-imposed coordination). We code coordination both as a continuous variable and as binary indicators for high-coordination (≥4) versus low-coordination (≤2). This allows us to interpret interaction effects.

Union density (ud): The Secondary moderator, is the share of employees who are union members. Union density captures bargaining power but not the institutional structure that determines how that power translates into outcomes.

Adjusted bargaining coverage (AdjCov): A benchmark moderator, defined as the percentage of workers whose wages are determined by collective agreements, capturing both direct union membership and extensions to non-unionised workers through legal mechanisms. Due to limited sample size, the variable is only used for certain robustness checks.

All three moderators are frozen at pre-period baseline values (averaged over a window prior to the panel start, e.g. 2000–2005) to treat institutions as time-invariant structural characteristics. This follows Leibrecht et al. (2023), who freeze union density and coordination at pre-sample values to prevent reverse causality as automation may itself induce institutional change. Freezing also eliminates within-country time-series variation, meaning the identifying variation for interaction terms is purely cross-country.

4.2.4 Macroeconomic Controls

GDP growth from XXX, is used as the primary macroeconomic control, following Leibrecht et al. (2023). GDP growth absorbs business-cycle fluctuations that simultaneously affect export demand and investment in robotization.

4.2.6 Summary of Control Variables

Variable	Source	Level	Time Fixed	Rationale
ln_va (log value added)	WIOD SEA	country-industry-year	Yes	Controls industry size/productivity
ln_cap (log capital compensation)	WIOD SEA	country-industry-year	Yes	Controls non-robot capital deepening
GDP growth	World Development Indicators	country-year	Yes	Absorbs business-cycle fluctuations
Country FE	n/a	country	n/a	Absorbs time-invariant country heterogeneity
Year FE	n/a	year	n/a	Absorbs common shocks
Industry FE	n/a	industry	n/a	Absorbs time-invariant industry heterogeneity

CLEAN UP

*Unemployment was originally planned as a control, but missing data across the 21 country panel made it not possible. (we may still look at including more controls). At the moment these are mainly sourced and rationalized from prior papers.

4.3 Sample Construction

4.3.1 Merging Procedure

The analytical panel is constructed through a sequential inner-merge of five secondary data sources, linked on country (ISO-2 code), industry (collapsed NACE Rev. 2 code), and year. The procedure proceeds in four stages. create visual?

> **Stage 1:** Robot intensity (IFR). The IFR robot-stock data arrive in fine-grained industry codes that do not map one-to-one to NACE Rev. 2 classifications. Multiple IFR codes are collapsed into 11 harmonised NACE manufacturing groups using a many-to-one crosswalk (e.g., IFR codes "10-12" → C10-C12; codes "26", "26-27", "27" → C26-C27). Where several IFR codes map to the same NACE group within a country-year, robot intensity is averaged across constituent codes XX. The data are filtered down to EU countries present in the IFR, manufacturing industries for years 2000–2014. The one-year lag of log robot intensity (ln_robots_lag1) is computed within each country-industry entity, which means the first usable analysis year becomes 2001. This filtering and aggregation step follows the approach of Graetz and Michaels (2018), who also collapse IFR industry codes to match their output data.
>
> **Stage 2**: Labour outcomes and industry controls. The WIOD SEA data base provides the dependent variable (H_EMPE, hours worked by employees) and two industry-level controls (VA_QI, real value added; K, capital stock) at a finer industry classification than the IFR. The 19 raw C-level manufacturing codes (e.g., C16, C17, C18) from WIOD SEA are aggregated

to the same 11 collapsed NACE groups via summation (e.g., C16 + C17 + C18 → C16-C18). All variables are log-transformed after aggregation. The merge with the Stage 1 IFR panel is an inner join on country-industry-year, retaining only observations present in both sources.

Stage 3: Institutional moderators (ICTWSS). Baseline-frozen institutional variables: coordination (coord_pre), union density (ud_pre), and adjusted bargaining coverage (adjcov_pre), are computed as country-level averages over 1990–1995 and merged on country code using a left join. Because these variables are time-invariant by construction, each country receives a single value that fills all its industry-year observations. Countries lacking a given moderator receive missing values for that variable but are not dropped from the panel at this stage. This follows XX.

Stage 4: Macroeconomic controls (Eurostat). GDP growth is computed as the log-difference of nominal GDP from Eurostat national accounts data and merged on country-year via a left join. Unemployment was originally planned as an additional control following Leibrecht et al. (2023), but a feasibility audit (See appendix XX) revealed missingness rates exceeding 10% across the 23-country panel, and it was dropped.

The resulting panel structure allows for fixed-effects estimation exploiting within-industry and within-country variation over time. XX

4.3.2 Final Sample Characteristics

The full analytical panel covers 23 EU countries, 11 collapsed NACE Rev. 2 manufacturing industries, and 14 analysis years (2001–2014). The countries retained after the IFR-WIOD SEA inner merge are: AT, BE, BG, CZ, DE, DK, EE, EL, ES, FI, FR, HU, IE, IT, LT, LV, NL, PL, PT, RO, SE, SI, SK. The 11 industries and their NACE codes are: C10-C12 (Food, beverages, tobacco), C13-C15 (Textiles, apparel, leather), C16-C18 (Wood, paper, printing), C19 (Coke, refined petroleum), C20-C21 (Chemicals, pharmaceuticals), C22-C23 (Rubber, plastics, non-metallic minerals), C24-C25 (Basic metals, fabricated metal), C26-C27 (Computer, electronic, electrical equipment), C28 (Machinery and equipment n.e.c.), C29-C30 (Motor vehicles, other transport), C31-C33 (Furniture, other manufacturing, repair).

UPDATED AS OF MARCH 26

Panel Data	
Countries	23
Rows	3960
Country-industry entities	264
Year range	2001-2014
Non-missing lagged robot observations	2283

The full sample for Eq. 1 XX contains approximately 226 country-industry entities and 2,212 observations. The panel is not perfectly balanced: not every country contributes all 11 industries across all 14 years, owing to variation in IFR reporting coverage across smaller EU member states. However, the degree of imbalance is modest, the vast majority of entities are observed for the full 14-year span, and two-way fixed effects estimation with an unbalanced panel remains valid provided that the missingness is not systematically correlated with the outcome.

The panel period (2001–2014) is determined by data availability rather than theoretical choice: 2000 is the first year of WIOD SEA coverage, and the one-year lag on robot intensity consumes 2000, making 2001 the first usable year. The terminal year 2014 reflects the end of the WIOD 2016 release cycle. This period conveniently excludes neither the initial diffusion period of industrial robots (which is well underway by 2001 in Western Europe) nor the post-2015 period where WIOD data are unavailable. XX

4.3.3 Sample Restrictions

The analysis uses four samples, each relating to a specific equation (or set of equations). The nesting structure creates a transparent view on exactly which countries and observations are lost at each restriction, and why. create visuals?

Full sample (Eq. 1, Eq. 2 coord). The broadest sample includes all 23 countries with non-missing values for the outcome (ln_h_empe), the lagged treatment (ln_robots_lag1), and the three industry-level controls (ln_va_wiod_qi, ln_k_wiod, gdp_growth). This yields approximately 226 entities and 2,212 observations. Eq. 1 (baseline) and Eq. 2 with coordination as the sole moderator are estimated on this sample, because coordination data from the ICTWSS are available for all 23 countries.

Joint coord-ud sample (Eq. 2b). The more exploratory, Equation 2b, requires non-missing values for both coordination and union density. Two countries, Romania (RO) and Slovakia (SK), lack pre-period union density data in the ICTWSS and are dropped, reducing the sample to 21 countries, 206 entities, and 2,068 observations. A diagnostic decomposition (See Appendix XX) confirms that the loss of these two countries does not drive any attenuation in the coordination interaction coefficient: re-estimating the Eq. 2 coord-only specification on the restricted 21-country sample produces a slightly larger coefficient than the full 23-country estimate. The attenuation in Eq. 2b is attributable to the addition of ud-related interaction terms, not to sample composition, as also shown in the diagnostic decomposition (see Appendix XX), as well as backed by previous literature such as Leibrecht et al. (2023)

Remove ADJCOV completely?

AdjCov restricted sample (Eq. 2 adjcov). Adjusted bargaining coverage has more limited availability in the ICTWSS, restricting the sample to a subset of countries that clear a lower threshold (≥14 countries). This sample is used only for the secondary moderation test and should be interpreted with caution given the reduced cluster count.

Four categories of countries have been removed from the final panel. First, countries not covered by the IFR at sufficient industry granularity (e.g., Cyprus, Malta, Luxembourg, Croatia) are dropped at Stage 1 because no robot-stock data exist at the manufacturing sub-industry level. Second, countries present in the IFR but absent from the WIOD SEA (or vice versa) are lost at the Stage 2 inner merge. Third, within the 23-country full sample, countries lacking specific institutional moderators are excluded only from the equations that require those moderators (Romania and Slovakia from Eq. 2b, additional countries from the AdjCov specification).

5. Methodology

5.1 Methodological Approach

xx

5.2 Measurement

5.2.1 Dependent Variable: Labor Input

The dependent variable is industry-level labour input, operationalised as the natural logarithm of total hours worked by persons engaged (H_EMPE) from the WIOD Socio-Economic Accounts (Timmer et al., 2015). Hours worked is preferred over headcount employment because it captures both the extensive margin (hiring and separations) and the intensive margin (adjustments to hours per worker), which are distinct channels through which robot adoption may affect labour demand (Graetz & Michaels, 2018). The log transformation serves two purposes: it normalises a right-skewed distribution across heterogeneous industries and allows coefficients to be interpreted as semi-elasticities, consistent with standard labour demand specifications.

5.2.2 Independent Variable: Robot Intensity

Robot intensity is constructed as the operational stock of industrial robots per thousand workers in country i, industry j, year t, using IFR data on robot stocks and a base-period employment denominator:

Create LATEX version

Robot Intensity_ijt = Robot Stock_ijt / Employment_ij,base

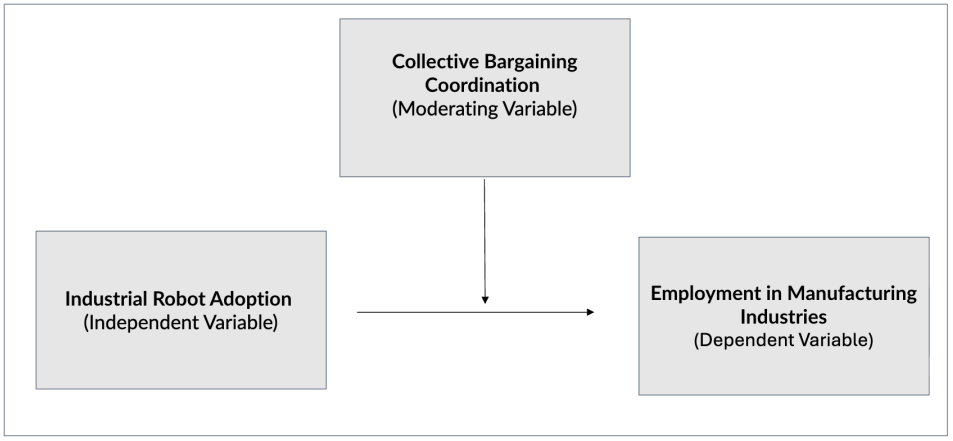
The denominator is fixed at a pre-sample baseline year rather than contemporaneous employment, following Leibrecht et al. (2023). The variable is transformed into a natural logarithm and lagged one period (t−1). The log transformation accommodates the highly skewed distribution of robot stocks across industries (e.g., automotive versus food processing).

5.2.3 Moderator Variables: Institutions

Clean up, alot of cross over with 4.0 now

Bargaining coordination (coord) is the primary institutional moderator, sourced from the ICTWSS database (Visser, 2021). The index ranges from 1 (fragmented firm-level bargaining) to 5 (centralised or government-imposed coordination) and captures the degree of horizontal synchronisation across bargaining units during wage-setting (Kenworthy, 2001).

Union density (ud), the share of employees who are union members, serves as a reference benchmark. It captures bargaining power but not the institutional architecture through which that power translates into adjustment outcomes.



Gunnarsson & Andersson (2026)

5.2.4 Control Variables

This is in the 4.0 data section now, perhaps move here, check w/ other papers

5.3 Empirical Specifications

5.3.1 Equation 1: Baseline Model

Equation 1 estimates the unconditional association between robot adoption and industry-level labour input across European manufacturing:

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$$\ln(H_EMPE)_ijt = \beta_1 \ln(Robots)_\{ij,t-1\} + \gamma_1 \ln(VA_QI)_\{ijt\} + \gamma_2 \ln(K)_\{ijt\} + \gamma_3 GDPGrowth_it + \alpha_ij + \delta_t + \varepsilon_ijt$$

where i indexes countries, j manufacturing industries, and t years. The coefficient β_1 captures the average within-entity semi-elasticity of labour input with respect to robot intensity, conditional on industry output, capital, macroeconomic conditions, and the full fixed-effects structure. The sample comprises approximately XX 21–23 EU manufacturing countries over 2000–2014.

This equation serves as a benchmark: if $\beta_1 \approx 0$, the average effect masks heterogeneity that the moderation models are designed to unpack. If $\beta_1 < 0$, robots are associated with labour displacement on average; if $\beta_1 > 0$, with labour complementarity. Following the prior literature (Graetz & Michaels, 2018), a near-zero or weakly negative baseline is expected.

5.3.2 Equation 2:

Equation 2 introduces the core moderation test by interacting robot intensity with the baseline-frozen, centred coordination index:

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$$\ln(H_EMPE)_ijt = \beta_1 \ln(Robots)_\{ij,t-1\} + \beta_2 [\ln(Robots)_\{ij,t-1\} \times coord_pre_c] + \gamma_1 \ln(VA_QI)_\{ijt\} + \gamma_2 \ln(K)_\{ijt\} + \gamma_3 GDPGrowth_it + \alpha_ij + \delta_t + \varepsilon_ijt$$

The standalone main effect of coord_pre_c does not appear because it is a country-level, time-invariant variable fully absorbed by α_ij. The coefficient of interest instead becomes β_2. Which captures whether a one-unit increase in baseline coordination changes the slope of robot adoption on labour input.
Because centering sets the moderator's mean to zero, β_1 is now interpretable as the robot-labour association for a country at the sample-mean level of coordination.
The marginal effect of robots on labour input at a given coordination level is $\beta_1 + \beta_2 \times$ coord_pre_c. This is reported graphically alongside confidence bands to convey how the association varies across the observed range of institutional configurations.

The same Equation 2 structure is estimated with two alternative moderators, union density (ud_pre_c) and adjusted bargaining coverage (adjcov_pre_c), as secondary and benchmark comparisons, respectively.

Because the moderators are country-level and time-invariant, the effective sample size for identifying β_2 is the number of countries, not the number of entity-year observations. With 21–23 countries, this places the design in a few-clusters setting where conventional clustered standard errors may under-reject. We address this with wild cluster bootstrap inference (see Section 5.3), and frame interaction results as suggestive evidence rather than definitive causal tests.

5.3.3 Equation 2b:

Equation 2b is an exploratory extension that simultaneously includes coordination and union density interactions, plus their three-way product:

$$\begin{aligned}\ln(H_EMPE)_ijt = {} & \beta_1 \ln(Robots)_\{ij,t-1\} \\ & + \beta_2 [\ln(Robots)_\{ij,t-1\} \times coord_pre_c] \\ & + \beta_3 [\ln(Robots)_\{ij,t-1\} \times ud_pre_c] \\ & + \beta_4 [\ln(Robots)_\{ij,t-1\} \times coord_pre_c \times ud_pre_c] \\ & + \gamma_1 \ln(VA_QI)_\{ijt\} + \gamma_2 \ln(K)_\{ijt\} + \gamma_3 GDPGrowth_it \\ & + \alpha_ij + \delta_t + \varepsilon_ijt\end{aligned}$$

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This specification tests whether the coordination moderation itself depends on union density, a hypothesis motivated by the possibility that coordination is only effective where unions retain sufficient bargaining power to enforce coordinated agreements. The three-way term β_4 captures this second-order interaction.

Equation 2b is exploratory for two reasons. First, it introduces three additional interaction terms identified off 21 country-level clusters, raising over-parameterisation concerns. Sample decomposition analysis confirms that the attenuation of the coordination coefficient in Eq. 2b (relative to Eq. 2) is driven by multicollinearity between the coord and ud interaction terms rather than by sample composition changes. Second, the theoretical prediction for the three-way term is underdetermined: it could be positive (coordination needs union power to function) or negative (coordination substitutes for union power).

6. Empirical Analysis

6.1 Main Results

6.1.1 Descriptive statistics

6.1.2 Baseline (Eq. 1)

6.1.3 Coordination moderation (Eq. 2) — primary

6.1.4 Density and adjusted coverage (Eq. 2 variants)

6.1.5 Three-way interaction (Eq. 2b) — exploratory

6.2 Robustness

multicollinearity test ⇐ run this test

6.2.1 Inference (wild cluster bootstrap)

6.2.2 Sample sensitivity (jackknife, common-sample, Switzerland)

6.2.3 Specification sensitivity (capital proxy, binary coord, robot stock)

6.2.4 Period sensitivity (2008–2009 exclusion)

6.2.5 Sample-vs-specification decomposition

7. Discussion

7.1 Theoretical implications — Hawk-Dove and the coordination mechanism

7.2 Empirical implications — relation to Graetz & Michaels (2018) and Leibrecht et al. (2023)

7.3 Sectoral implications — what coordination means for manufacturing-exposed industries

7.4 Policy implications — bargaining institutions as automation-management infrastructure

8. Limitations

8.1 Data limitations (WIOD/IFR/ICTWSS measurement; baseline window; coordination as ordinal)

8.2 Methodological limitations (TWFE not IV; few clusters at lower bound; bad-control risk)

8.3 Scope limitations (manufacturing only; 21-country gap to Leibrecht's 37; 2001–2014 window)

8.4 Avenues for future research

9. Conclusion

APPENDIX

APPENDIX 1: Game Theory (Related to Section 3.2 The Hawk-Dove Divide)

To formalize a union's willingness to engage in conflict, a symmetric Hawk-Dove two world game can be constructed (Figure Z). Within each world two sector-level unions simultaneously choose between an aggressive Hawk strategy (demanding high wages, resisting automation-driven restructuring) and a Dove strategy (exercising wage restraint and accepting technological adjustment).

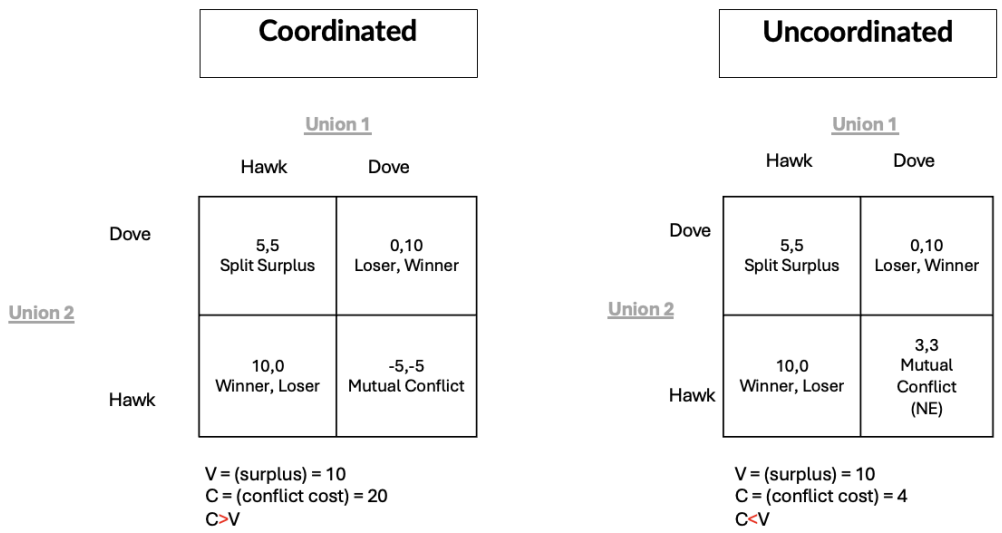

Figure Z: Hawk–Dove Game under Coordinated and Uncoordinated Collective Bargaining

Let V denote the economic surplus generated by automation-driven productivity gains and C the cost of bargaining conflict (e.g., lost output, strikes, or competitiveness losses). A Dove-Dove choice splits the surplus equally (union_1D,2D=V/2). A single Hawk captures the full surplus at the other's expense (union_1H=V & union_2D=0 or union_2H=V & union_1D=0). Lastly, a mutually aggressive Hawk-Hawk stance yields the works (V−C)/2 outcome (union_1H,2H=(V−C)/2).

The critical insight is that surplus V depends on whether workers across unions are complements or substitutes. In coordinated systems, workers are interdependent: aggression in one sector destroys output in others, raising C above V and making Hawk-Hawk yield negative payoffs, shifting the equilibrium toward Dove/Dove. In uncoordinated systems, workers are substitutes: aggression redirects demand rather than destroying it, keeping C low and sustaining Hawk as a dominant strategy, locking both unions into mutual Hawk/Hawk conflict. Coordination becomes the key mechanism that raises C by forcing unions to internalize cross-sector externalities, shifting behaviour from Hawkish to Dovish.

APPENDIX 2: Coordination Interaction (Related to Section XXXX)

*CLEAN UP - AS OF MARCH 26th

To ensure that the attenuation of the coordination interaction in Eq. 2b is not driven by the loss of two countries (Romania and Slovakia lack ud data), we re-estimate the Eq. 2 coord-only specification on the exact 21-country Eq. 2b sample. The coord coefficient is slightly larger (0.013 vs. 0.011) and marginally significant under wild cluster bootstrap (p = 0.095 vs. 0.121), indicating that the two dropped countries were not carrying the Eq. 2 result. The attenuation in Eq. 2b (0.013 ⇒ 0.009, p_wild = 0.57) is therefore attributable to the addition of ud-related interaction terms, which share within-panel variation with the coord interaction in a setting with only 21 clusters. This is consistent with the interpretation that coordination is the empirically active moderator and that adding union density as a second moderator introduces multicollinearity without informational gain.

WIOD Eq. 2 Coordination — Sample vs. Specification Decomposition

Model	Sample	N_countries	N_entities	N_obs	Coef	SE	p_country	p_wild
Eq. 2 coord (full)	23-country full	23	226	2,212	0.0108	0.0046	0.019	0.121
Eq. 2 coord (Eq. 2b sample)	21-country intersection	21	206	2,068	0.0128	0.0044	0.004	0.095
Eq. 2b coord term	21-country intersection	21	206	2,068	0.0088	0.0070	0.210	0.568

Decomposition of the attenuation (full Eq. 2 → Eq. 2b):

Step	Comparison	Δ Coef	Δ p_wild	Interpretation
Sample effect	Full → 21-country, coord-only	+0.0020	−0.025	Two dropped countries diluted the result; no attenuation
Specification effect	21-country coord-only → Eq. 2b	−0.0040	+0.472	Adding ud terms absorbs shared variance; drives entire attenuation
Total	Full Eq. 2 → Eq. 2b	−0.0020	+0.447	Net: modest coefficient drop, large p-value collapse